Year 13 Further Maths - Centres of mass

Similar questions - Solids of revolution and laminas under curves

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The region bounded by the x-axis, the curve $y = \sqrt{3x+1}$ and the lines $x = 0$ and $x = 3$ is rotated through 2π about the x-axis to form a uniform solid R. The centre of mass of R is G.

(a) Explain why G lies on the x-axis. **[1]**

(b) Show that the x-coordinate of G is $\frac{21}{11}$. **[4]**

(c) R is freely suspended from the point P on the rim of the curved surface where $x = 3$ and hangs in equilibrium. Find the angle between the axis of symmetry and the vertical. **[4]**

2 A uniform lamina L occupies the region bounded by the x-axis, the y-axis, the line $x = 4$ and the curve $y = 3 + \cos x$ (x in radians). The centre of mass of L is $(\bar{x}, \bar{y})$.

(a) Show that $\bar{x} = 1.72$ correct to 3 s.f. **[4]**

(b) Find $\bar{y}$, correct to 3 s.f. **[3]**

(c) L is freely suspended from the corner at $(4,0)$. Find the angle that the edge along the x-axis makes with the vertical. **[3]**

3 In this question you must show detailed reasoning. A uniform lamina occupies the region bounded by the axes, the line $x = 3$ and the curve $y = \dfrac{8}{\sqrt{x^2+4}}$. Its centre of mass is $(\bar{x}, \bar{y})$.

(a) Show that $\bar{x} = \dfrac{\sqrt{13}-2}{\sinh^{-1} 1.5}$. **[4]**

(b) Show that $\bar{y} = \dfrac{2\tan^{-1} 1.5}{\sinh^{-1} 1.5}$ and evaluate it to 3 s.f. **[4]**

(c) The lamina is freely suspended from the point $(3,0)$. Find the acute angle between the x-axis edge and the horizontal. **[3]**

Question 1

(a) The solid is symmetric about the x-axis, so $\bar{y} = 0$

(b) Volume $= \pi \int_0^3 (3x+1)\mathrm{d}x = \pi \left[\frac{3}{2}x^2 + x\right]_0^3 = 16.5\pi$. $\int_0^3 x(3x+1)\mathrm{d}x = \left[x^3 + \frac{1}{2}x^2\right]_0^3 = 31.5$. $\bar{x} = \dfrac{31.5}{16.5} = \dfrac{21}{11} = 1.91$

(c) Radius at $x = 3$ is $\sqrt{10}$. Angle between PG and the axis: $\tan\phi = \dfrac{\sqrt{10}}{3 - 21/11} = \dfrac{\sqrt{10}}{12/11}$, $\phi = 71.0^\circ$

Question 2

(a) Area $= \int_0^4 (3+\cos x)\mathrm{d}x = 12 + \sin 4 = 11.243$. $\int_0^4 x(3+\cos x)\mathrm{d}x = \left[\frac{3}{2}x^2 + x\sin x + \cos x\right]_0^4 = 24 + 4\sin 4 + \cos 4 - 1 = 19.32$. $\bar{x} = \dfrac{19.32}{11.243} = 1.72$

(b) $\bar{y} = \dfrac{1}{A}\int_0^4 \frac{1}{2}(3+\cos x)^2\mathrm{d}x = \dfrac{1}{11.243}\left[\frac{19}{4}x + 3\sin x + \frac{1}{8}\sin 2x\right]_0^4 = \dfrac{16.85}{11.243} = 1.50$

(c) G relative to $(4,0)$ is $(-2.28, 1.50)$; PG is vertical, and the edge along the x-axis is in direction $(-1,0)$, so the angle is $\tan^{-1}\dfrac{1.50}{2.28} = 33.3^\circ$

Question 3

(a) Area $= 8\left[\sinh^{-1}\frac{x}{2}\right]_0^3 = 8\sinh^{-1}1.5$. $\int_0^3 \dfrac{8x}{\sqrt{x^2+4}}\mathrm{d}x = 8\left[\sqrt{x^2+4}\right]_0^3 = 8(\sqrt{13} - 2)$. So $\bar{x} = \dfrac{\sqrt{13} - 2}{\sinh^{-1}1.5} = 1.34$

(b) $\int_0^3 \frac{1}{2}\cdot\dfrac{64}{x^2+4}\mathrm{d}x = 32\cdot\frac{1}{2}\left[\tan^{-1}\frac{x}{2}\right]_0^3 = 16\tan^{-1}1.5$. $\bar{y} = \dfrac{16\tan^{-1}1.5}{8\sinh^{-1}1.5} = \dfrac{2\tan^{-1}1.5}{\sinh^{-1}1.5} = 1.65$

(c) G relative to $(3,0)$ is $(-1.66, 1.65)$; the edge from $(3,0)$ along the x-axis has direction $(-1,0)$. Angle $= \tan^{-1}\dfrac{1.645}{1.656} = 44.8^\circ$ with the vertical (the angle with the horizontal is $90^\circ - 44.8^\circ = 45.2^\circ$)